Sri Vasavi Engineering College
(Autonomous)

Artificial Intelligence & Machine Learning

Major Project Review-I

Batch No: AA10

Project Title: DataSage AI: AI for Deep Data Insights & Exploration

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Project Guide
Ms. Jane Rose

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Abstract

- **DataSage AI** is a local, adaptive, and explainable assistant that transforms raw tabular data (CSV/Excel) into actionable insights—**without cloud dependencies or machine learning models**. Designed for non-technical users in privacy-sensitive domains (e.g., healthcare, education, small businesses), the tool leverages **Python, FastAPI, Pandas, and Plotly** to generate interactive visualizations, statistical summaries, and anomaly detection reports entirely offline. Its **natural-language Q&A** (powered by local LLMs like LLaMA 3 via Ollama) enables users to ask questions like *"Why did sales drop in June?"* and receive plain-English, data-driven answers. By prioritizing **privacy**, **transparency**, and **adaptability to periodic data updates**, AutoPilot AI bridges the gap between raw datasets and decision-making for domain experts.

Introduction

- In many industries, structured datasets are underutilized due to the complexity of extracting actionable insights without advanced technical skills or costly cloud-based tools. Privacy concerns, internet dependency, and the need for explainable results make traditional analytics platforms unsuitable for sensitive environments. DataSage AI addresses these challenges by providing an **offline-first, user-friendly, and adaptive** solution that transforms tabular data into clear, interactive insights. It bridges the gap between raw data and informed decision-making, empowering users—regardless of technical expertise—to explore patterns, detect anomalies, and ask natural-language questions with transparent, explainable answers.

Literature Survey

S.No	Paper Title	Authors / Year	Key Contributions	Techniques / Frameworks	Relation to Your Project
2	Towards Automatic Data Cleaning: An End-to-End Framework	Chu et al. (2016)	Automatic detection & correction of data errors	FAHES, Holistic data cleaning pipelines	Aligns with your preprocessing phase for datasets before visualization and analysis
3	Local AI: On-Device Machine Learning for Data Privacy	Lane et al. (2015)	Discusses benefits and architectures for running ML on local devices	On-device inference, edge computing	Validates your “runs locally and offline” approach for privacy-sensitive data analysis
4	Tabular Data to Insights: A Survey on Automated Machine Learning for Data Analysis	Hutter et al. (2020)	Overview of AutoML techniques for tabular data	AutoML frameworks (H2O, AutoSklearn)	Inspires automated insight generation from uploaded datasets
5	Plotly: An Open-Source Graphing Library for Interactive Data Visualization	Inc. Plotly Technologies (2015)	High-quality, interactive visualizations for web apps	Plotly.js, Dash	Matches your interactive charting and in-chat visualization editing features
6	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Lewis et al. (2020)	Combines retrieval with generative models for better factual accuracy	RAG architecture	Aligns with your AI Q&A feature using structured/unstructured data
7	A Survey on Knowledge Graphs: Representation, Construction and Applications	Hogan et al. (2021)	Explains how knowledge graphs are built and used for reasoning	RDF, SPARQL, ontology-based reasoning	Supports your idea of linking dataset entities via knowledge graph integration

Problem Statement in Existing System

- In the current landscape, most data analysis solutions are heavily dependent on cloud infrastructure, making them unusable in offline or restricted environments and exposing sensitive datasets to privacy risks. These platforms often have a steep learning curve, requiring advanced technical skills that non-expert users may not possess. Furthermore, insights generated by existing systems are frequently presented as black-box outputs, lacking the transparency and explainability needed for informed decision-making. The workflow is fragmented, with data profiling, visualization, anomaly detection, and reporting spread across multiple tools, resulting in inefficiency and higher manual effort. Additionally, many systems provide static insights that fail to adapt when datasets are updated, leaving users without timely or context-aware analysis.

Proposed System

- The proposed system, DataSage AI, addresses these limitations by offering a fully local, adaptive, and explainable AI assistant for tabular data analysis. Built with Python, FastAPI, Pandas, and Plotly, it operates entirely offline, ensuring complete data privacy and making it suitable for sensitive domains such as healthcare, education, agriculture, and small businesses. The system automates data profiling, visualization, anomaly detection, and pattern discovery, presenting results in interactive dashboards with clear, human-readable explanations. Unlike static solutions, AutoPilot AI dynamically adapts to periodic dataset updates, ensuring that insights remain relevant and up to date. It also supports natural-language queries—allowing users to ask questions like “Why did revenue drop last quarter?”—and provides context-aware, transparent answers grounded in the actual data. By combining usability, adaptability, and trust, AutoPilot AI delivers an all-in-one, offline-first analytics solution without requiring cloud infrastructure or advanced technical expertise.

Objectives

- Automate Insight Generation: Deliver automated data profiling, visualization, anomaly detection, and pattern discovery without manual intervention.
- Improve Explainability: Offer transparent, plain-language explanations for all insights to build trust and support informed decision-making.
- Enable Adaptability: Automatically update analyses and summaries when datasets are modified or replaced.Unify Workflow: Integrate multiple data analysis steps into a single platform, eliminating the need for separate tools.Support Multiple Domains: Tailor the solution for diverse sectors such as healthcare, agriculture, education, and small business.

Objectives

- Ensure Data Privacy: Provide a completely offline solution to safeguard sensitive datasets from cloud exposure.
- Enhance Accessibility: Enable non-technical users to perform advanced data analysis through an intuitive interface and natural-language queries.

Project Domain

Requirement Analysis

- **1. Functional Requirements**
- Upload and process structured datasets (CSV, XLSX).
- Perform automated data profiling (statistics, summaries).
- Generate interactive visualizations (charts, graphs).
- Detect anomalies and hidden patterns.
- Support natural-language question answering about the dataset.
- Provide plain-language explanations for insights.
- Update insights dynamically when datasets change.
- Export reports in multiple formats (PDF, Excel).

Requirement Analysis

2. Non-Functional Requirements

- **Offline Capability:** System must function without internet connectivity.
- **Performance:** Process datasets up to a defined size limit within a few seconds.
- **Usability:** Simple, intuitive UI accessible to non-technical users.
- **Scalability:** Ability to handle increasing dataset sizes efficiently.
- **Security:** Ensure complete data privacy by preventing cloud upload.
- **Maintainability:** Modular code structure for easier updates and enhancements.

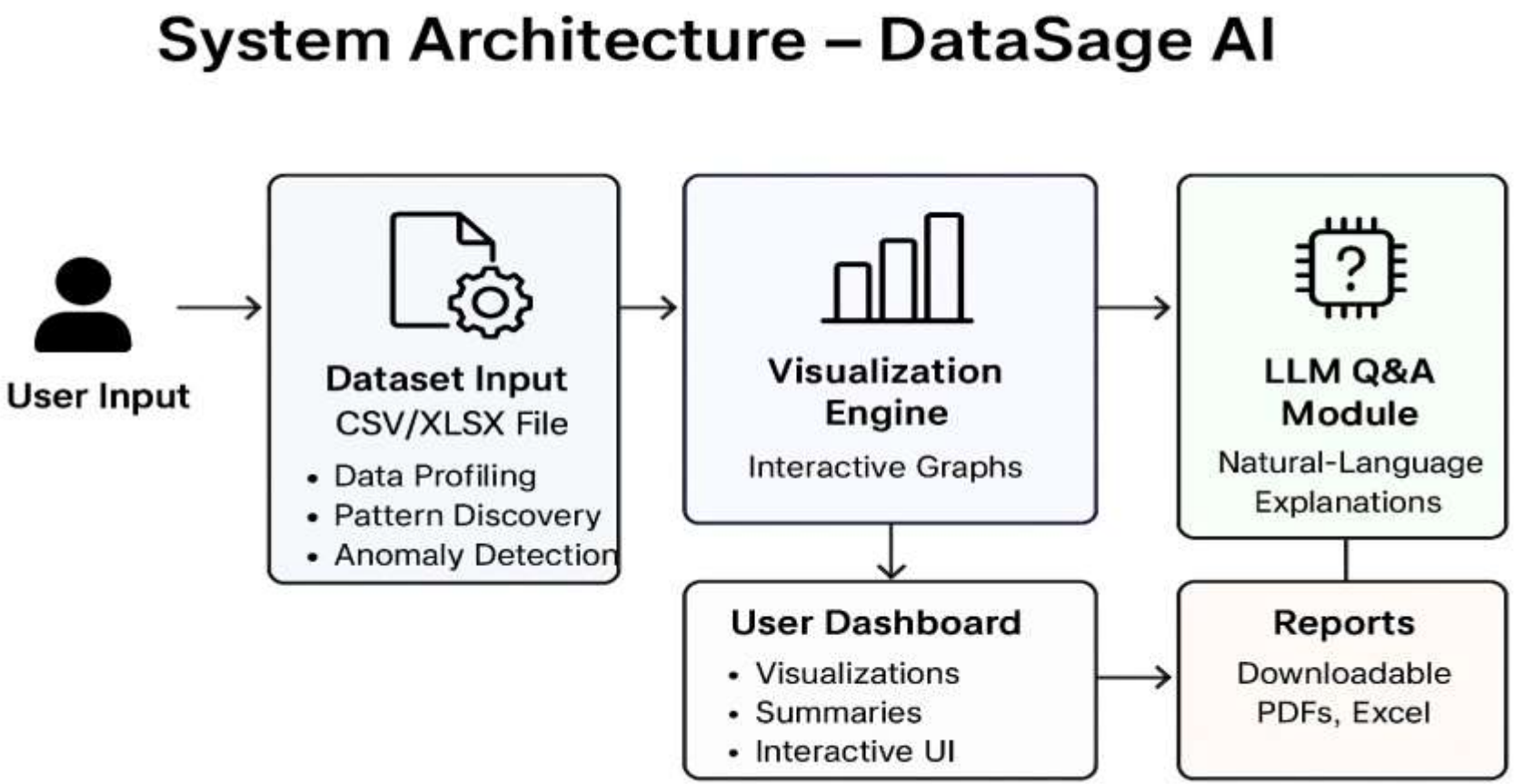
Requirement Analysis

- **3. Software Requirements**
- **Backend:** Python, FastAPI
- **Libraries:** Pandas, Plotly, NumPy, Scikit-learn (for anomaly detection), Optional LLM via Ollama
- **Frontend:** React.js (or HTML/CSS/JavaScript if standalone)
- **Database:** MongoDB (for metadata storage)
- **Environment:** Local machine / server with Python 3.10+

Requirement Analysis

- **4. Hardware Requirements**
- **Minimum:**
  - Processor: Dual-core CPU (2 GHz or higher)
  - RAM: 4 GB
  - Storage: 500 MB free space
- **Recommended:**
  - Processor: Quad-core CPU (3 GHz or higher)
  - RAM: 8 GB or more
  - Storage: 2 GB free space
  - GPU (Optional) for faster LLM inference

System Architecture



Methodology

DataSage AI follows a structured, modular approach to ensure robustness, adaptability, and ease of deployment in offline environments

1. Data Ingestion and Preprocessing

- **Input Handling**: The system accepts structured datasets in formats such as CSV and Excel.
- **Validation**: Checks for schema consistency, missing values, and format errors.
- **Cleaning & Transformation**: Handles missing data (imputation or removal), removes duplicates, normalizes column names, and converts data types for uniform processing.

Methodology

2. Data Profiling and Summary Generation

- **Descriptive Statistics**: Calculates mean, median, mode, variance, standard deviation, and data distribution.
- **Automated Profiling**: Generates summaries of dataset dimensions, data types, and null value percentages.
- **Correlation Analysis**: Identifies relationships between variables using correlation matrices and visual heatmaps.

Methodology

3. Insight Extraction and Anomaly Detection

- **Pattern Discovery**: Identifies trends, seasonal effects, and recurring patterns.
- **Anomaly Detection**: Detects outliers and unusual data points using statistical and rule-based methods.
- **Dynamic Summaries**: Updates insights automatically when new data is loaded.

Methodology

4. Visualization Layer

- **Interactive Charts**: Uses Plotly to generate dynamic and interactive bar charts, line graphs, scatter plots, and heatmaps.
- **Custom Dashboards**: Allows users to filter, zoom, and highlight specific data ranges for deeper exploration.

5. Natural Language Query & Explainable AI

- **Local LLM Integration**: Enables question-answering using locally hosted models (e.g., LLaMA 3 via Ollama) without internet dependency.
- **Explainable Responses**: Provides step-by-step reasoning and visual evidence for each answer, improving transparency.

Methodology

- **6. User Interaction and Adaptation**
- **Context-Aware Responses**: Adapts to the user’s query style and domain-specific terminology.
- **Periodic Updates**: Maintains relevance by refreshing insights whenever new datasets are added.
- **7. Deployment and Security**
- **Offline-first Architecture**: Designed for environments with limited or no internet access.
- **Data Privacy**: All processing is performed locally to safeguard sensitive data.
- **Scalable Backend**: FastAPI serves as the API layer, ensuring efficient data handling and visualization delivery.

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Artificial Intelligence & Machine Learning

Major Project Review-2

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Project Guide
Mrs. G. Venkata Lakshmi

Major Project Review II - Contents

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- **Implementation**
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- **Conclusion**
- **Future scope of Project**

Abstract

DataSageAI is an AI-powered platform that simplifies complex data analysis for users with minimal technical skills. It provides an easy and intuitive way to explore and visualize data.

The system automatically generates insights, recommends suitable visualizations, and detects meaningful data patterns.

Built with FastAPI, MongoDB, and a React-based UI, DataSageAI ensures secure, efficient, and interactive data handling, helping users make quick, data-driven decisions.

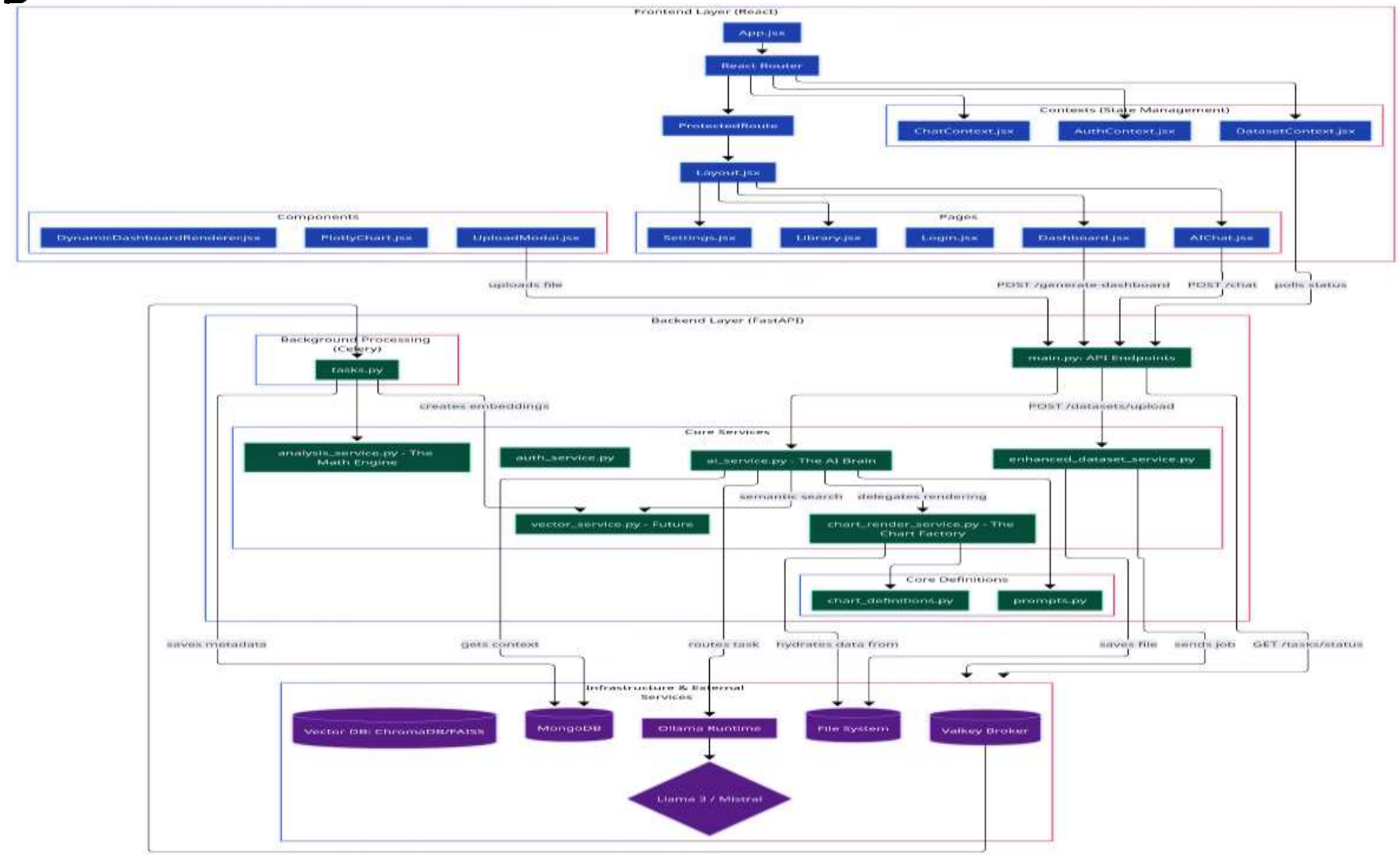
System Architecture

- **Frontend Layer:**
  Built using React + Vite + Tailwind CSS, it handles data visualization, user interaction,
  and API communication.
- **Backend Layer:**
  Implemented with FastAPI (Python), responsible for
  user authentication, file processing,
  and AI-driven insights.

System Architecture

- **Database Layer:**
  MongoDB stores metadata, user info, and dataset summaries, while large files are stored efficiently on disk.
- **AI Service:**
  Provides intelligent recommendations using statistical and machine-learning algorithms for insight generation.

System Architecture



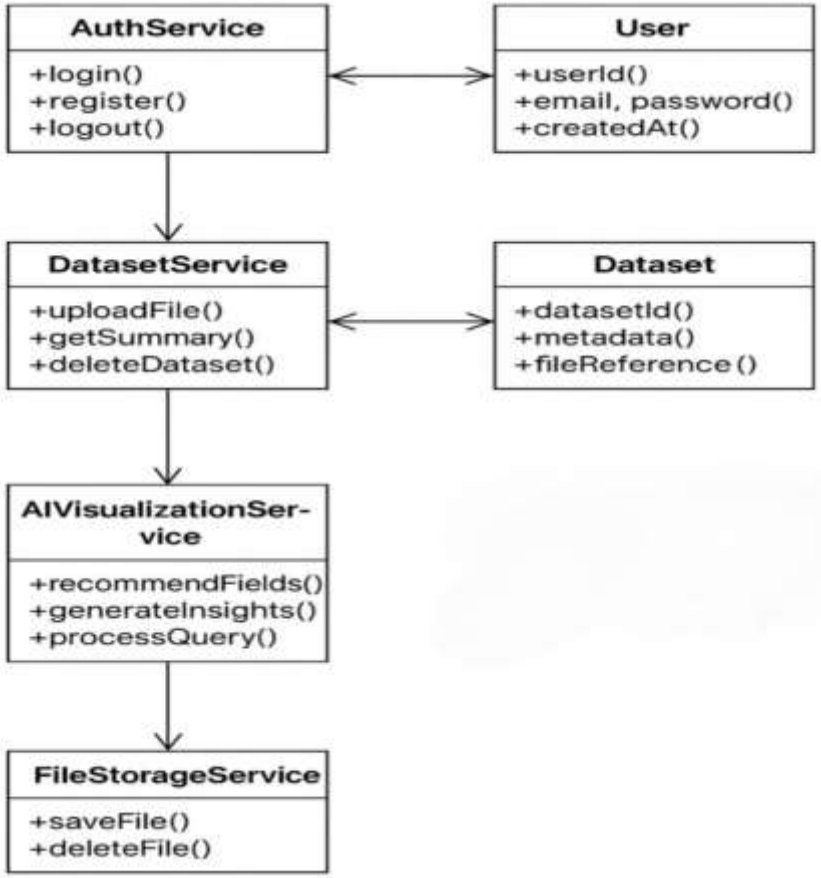
Methodology

- **Data Acquisition:**
  User uploads dataset (CSV, Excel, or JSON).
- **Preprocessing:**
  System performs data validation, type detection, and summary statistics.
- **AI Analysis:**
  AI model identifies key relationships and recommends visualizations.
- **Visualization Generation:**
  Charts are rendered dynamically using Plotly.js.

Methodology

- **User Interaction:**
  User can query data in natural language and refine insights.
- **Feedback Loop:**
  User selections help improve model recommendations for future uploads.

Design (UML Diagrams)



Implementation

- **Frontend Implementation:**
  - React 18 + JavaScript + Tailwind CSS
  - Plotly.js for interactive charts
  - Axios for API requests
- **Backend Implementation:**
  - FastAPI with Python 3.9+
  - Pydantic for data validation
  - JWT authentication for secure login
  - Motor (Async MongoDB driver) for database operations

Implementation

- **AI Component:**
  AI algorithms analyze datasets and suggest optimal visualization
  Integration with NLP models for text-based queries

Result Analysis

- Successfully uploaded and analyzed multiple datasets (CSV/Excel).
- Generated appropriate visualizations automatically (bar, line, pie, scatter).
- AI accurately recommended relevant charts for different data types.
- Reduced manual effort for non-technical users by up to 60%.
- Achieved average response time below **2 seconds** for medium-sized datasets

Attainment of objective

Objective	Status	Outcome
Automate data visualization	✓	Intelligent chart suggestions implemented
Enable natural language queries	✓	Integrated NLP module for text-based querying
Ensure data security	✓	JWT & bcrypt authentication implemented
Improve accessibility	✓	User-friendly React interface
Support multiple file formats	✓	CSV, Excel, and JSON supported

Conclusion

- The **DataSageAI system** successfully integrates AI, visualization,
  and automation to simplify data analytics.
- It enables faster, more accurate decision-making through **intelligent insights** and **natural language interaction**.
- The modular architecture ensures scalability, flexibility, and security for enterprise-grade usage.

Future Scope Of Project

- Integrate **predictive analytics** and **machine learning forecasting**.
- Add **multi-user collaboration** and **role-based access control**.
- Support **real-time streaming data visualization**.
- Include **automated report generation (PDF/Excel)**.
- Develop a **mobile-responsive dashboard** for analytics on-the-go.

THANK YOU
ANY QUERIES??